

8 (**short**) The data below describe the oxidation of a steel at 1220 °C in an atmosphere containing oxygen.

Time, t (s)	500	1000	2000	5000	10000
Mass gain per unit area, Δm (g m ⁻²)	4.5	6.0	8.5	13.5	19.5

(a) Demonstrate that, at 1220 °C, the kinetics of oxidation are approximately parabolic such that $(\Delta m)^2 = k t$ and that the parabolic rate constant k is about 0.038 g²m⁻⁴s⁻¹. [4]

(b) Assuming the oxide has the composition Fe₃O₄, could a sheet of this steel of thickness 1 mm, totally immersed in this environment, be used for 300 days without being oxidised throughout its entire thickness?

Relative atomic weight of oxygen = 16.

Relative atomic weight of iron = 56.

Density of the steel = 7800 kg m⁻³. [6]